Introduction to Earth Science

Unit 3: The Water Cycle

These sample lecture notes were written to try out Exam Forge. The content is for demonstration only.

1. Stages of the water cycle

- **Evaporation:** Water on the surface of oceans, lakes and rivers turns into water vapor using energy from the Sun. About 86% of the moisture entering the atmosphere comes from the oceans.
- **Transpiration:** Plants release water vapor through pores in their leaves called stomata. Together with evaporation this is called *evapotranspiration*.
- **Condensation:** Rising water vapor cools, reaches the dew point and becomes the water droplets that form clouds.
- **Precipitation:** Droplets in clouds merge, grow heavy and fall to the ground as rain, snow or hail.
- **Runoff and infiltration:** Part of the precipitation flows over the surface into streams, and part soaks into the soil and joins the groundwater.

2. Distribution of water

Reservoir	Share of all water
Oceans	96.5%
Glaciers and snow cover	1.7%
Groundwater	1.7%
Lakes, rivers, atmosphere	less than 0.1%

Most fresh water is held in glaciers. The fresh water people can use directly is a very small part of all water.

3. Residence time

The average time a water molecule stays in a reservoir is called its **residence time**. Water stays in the atmosphere for about 9 days on average, while in the oceans this reaches thousands of years. In deep groundwater, residence time can reach tens of thousands of years.

Summary: The water cycle is driven by solar energy and gravity. Solar energy lifts water into the atmosphere; gravity brings it back to the land and the seas through precipitation and runoff.